(3.) $f(x) = \begin{cases} x^2+1, & x>0 \\ 1, & x=0 \\ 1+x, & x<0 \end{cases}$

<u>LHL</u> $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} 1+x$

$= 1+0 = 1$

<u>RHL</u> $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x^2+1$

$= 0^2+1 = 1$

LHL = RHL

$\therefore \lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = 1$

$\therefore \lim_{x\to 0} f(x) = 1$

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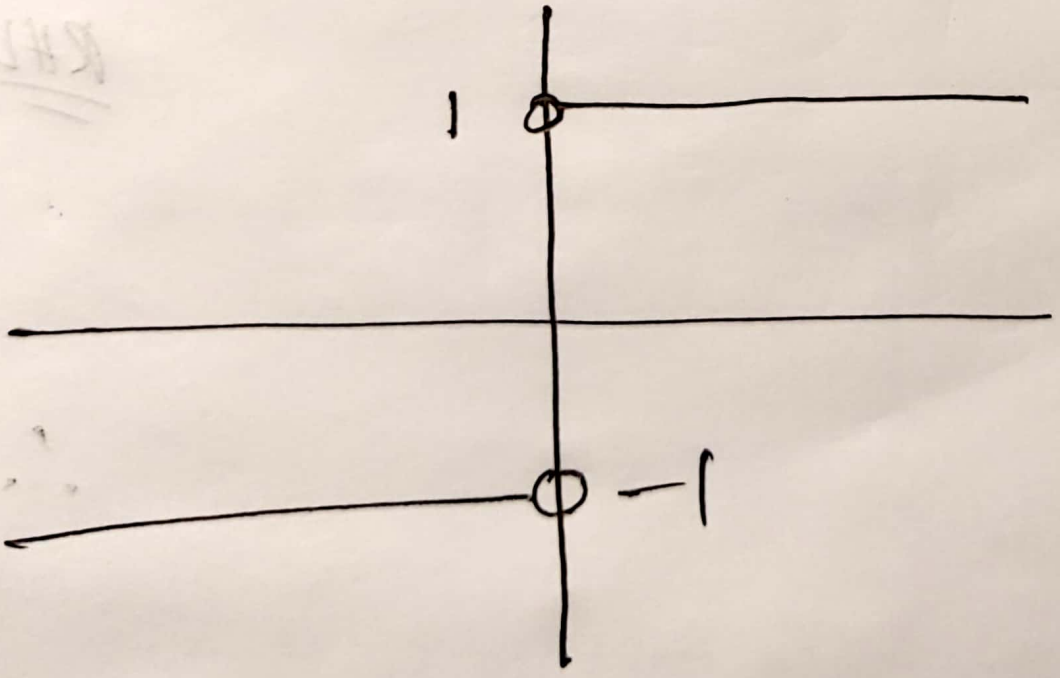
$$\lim_{x \to 0} \frac{x}{|x|} \Rightarrow \boxed{\frac{x}{|x|} = \begin{cases} \frac{x}{-x}, & x<0 \\ \frac{x}{x}, & x>0 \end{cases}}$$

<u>LHL</u> $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \frac{x}{-x} = -1$

<u>RHL</u> $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{x}{x} = 1$

LHL $\neq$ RHL

$\therefore$ $\lim_{x \to 0}$ doesnot exists



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$$\lim_{x \to \infty} \frac{3x+5}{6x+8}$$

$$\Rightarrow \lim_{x \to \infty} \frac{\cancel{x}\left(3+\frac{5}{x}\right)}{\cancel{x}\left(6+\frac{8}{x}\right)}$$

$$= \frac{3}{6} = \frac{1}{2}$$

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$$f(x) = \begin{cases} \frac{1}{x+2}, & x < -2 \\ x^2 - 5, & -2 < x < 3 \\ \sqrt{x+13}, & x > 3 \end{cases}$$

For $x = -2$,

LHL $\lim_{x \to -2^-} f(x) = \lim_{x \to -2^-} \frac{1}{x+2} = -\infty$

RHL $\lim_{x \to -2^+} f(x) = \lim_{x \to -2^+} x^2 - 5 = 4 - 5 = 1$

$\therefore \lim_{x \to 2} f(x)$ does not exist

For $x = 3$

LHL $\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} x^2 - 5 = 9 - 5 = 4$

RHL $\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} \sqrt{x+13} = \sqrt{3+13} = 4$

LHL = RHL = 4

$\therefore \lim_{x \to 3} f(x) = 4$

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Prove that $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e$

$$= \lim_{x \to \infty} \left(1 + x \cdot \left(\frac{1}{x}\right) + \frac{x(x-1)}{2!} \left(\frac{1}{x}\right)^2 + \frac{x(x-1)(x-2)}{3!} \left(\frac{1}{x}\right)^3 + \cdots \right)$$

$$= \lim_{x \to \infty} \left(1 + 1 + \frac{\cancel{x^2}\left(1 - \frac{1}{x}\right)}{2!} \frac{1}{\cancel{x^2}} + \frac{\cancel{x^3}\left(1 - \frac{1}{x}\right)\left(1 - \frac{2}{x}\right)}{3!} \frac{1}{\cancel{x^3}} + \cdots \right)$$

$$= 2 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots = e.$$

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